STAT5080 – HW 8 Solutions

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Problem 1

Two-way complete model with $a = 4$ levels of factor A and $b = 6$ levels of factor B, giving $ab = 24$ treatment combinations.

Part (a): Sample Size via Bonferroni and Tukey's Approach

We want simultaneous 95% CIs for all pairwise comparisons among the $ab = 24$ treatment combinations, each with length at most 6 (half-width $d = 3$). There are $m = \binom{24}{2} = 276$ pairwise comparisons.

Bonferroni half-width: $t_{ab(n-1),\,1-\alpha/(2m)} \times \sqrt{2\sigma^2/n}$.

Tukey's half-width: $\frac{q_{\alpha,\,ab,\,ab(n-1)}}{\sqrt{2}} \times \sqrt{2\sigma^2/n}$.

We iterate over n until the half-width $\leq d = 3$.

```
## Number of treatment combinations: ab = 24
## Number of pairwise comparisons:   m  = 276
##
## --- Bonferroni (last 5 rows) ---
##   n  v2   t_val     MSD     Action
##  34 792 3.76176 3.16051 Increase n
##  35 816 3.76123 3.11460 Increase n
##  36 840 3.76074 3.07063 Increase n
##  37 864 3.76027 3.02847 Increase n
##  38 888 3.75983 2.98801       STOP
##
## Bonferroni: n* = 38   Total N = 912
##
## --- Tukey's (last 5 rows) ---
##   n  v2   q_val     MSD     Action
##  32 744 5.16352 3.16200 Increase n
##  33 768 5.16291 3.11335 Increase n
##  34 792 5.16233 3.06688 Increase n
##  35 816 5.16178 3.02243 Increase n
##  36 840 5.16127 2.97986       STOP
##
## Tukey's:    n* = 36   Total N = 864
```

Bonferroni: The required sample size per treatment combination is $n^* = 38$, giving a total of $N = 24 \times 38 = 912$ observations.

Tukey's: The required sample size per treatment combination is $n^* = 36$, giving a total of $N = 24 \times 36 = 864$ observations.

Does the result make sense?

Yes. Tukey's method is specifically designed for all-pairwise comparisons and is less conservative than Bonferroni when m is large. With $m = 276$ comparisons, Bonferroni's per-comparison α correction is very stringent ($\alpha/2m \approx 0.000091$), inflating the required t critical value and thus requiring a larger sample size. Tukey's studentized range handles the same set of comparisons with a tighter critical value, so it requires fewer observations for the same CI width.

Part (b): Power-Based Sample Size

We apply the power method separately for each factor and take the maximum. For factor A ($a = 4$ levels, $\Delta_A = 4$, $P_A = 0.95$) and factor B ($b = 6$ levels, $\Delta_B = 6$, $P_B = 0.95$):

- Observations per level of A in the two-way layout: bn.
- Observations per level of B in the two-way layout: an.
- Error d.f.: $v_2 = ab(n-1)$.

Non-centrality parameter (least-favorable configuration, two means differ by Δ):

$$\lambda = \frac{(bn)\cdot\Delta_A^2/(2a)\cdot(a-1)}{\sigma^2} \quad \text{for factor } A, \quad \lambda = \frac{(an)\cdot\Delta_B^2/(2b)\cdot(b-1)}{\sigma^2} \quad \text{for factor } B.$$

```
## --- Factor A (delta=4, a=4, power >= 0.95) ---
##  n n_per_lev   Power     Action
##  2        12 0.57615 Increase n
##  3        18 0.80618 Increase n
##  4        24 0.91980 Increase n
##  5        30 0.96951       STOP
##
## Factor A requires n_A = 5 rep(s) per treatment combination.
##
## --- Factor B (delta=6, b=6, power >= 0.95) ---
##  n n_per_lev   Power     Action
##  2         8 0.65665 Increase n
##  3        12 0.89135 Increase n
##  4        16 0.97100       STOP
##
## Factor B requires n_B = 4 rep(s) per treatment combination.
```

$n^* = \max(5, 4) = 5$ replications per treatment combination. Total sample size: $N = 4 \times 6 \times 5 = 120$.

Problem 2

Two-way fixed effects model with $a = 5$ levels of factor A and $b = 6$ levels of factor B. We use **Method 2 (CI-based)**: find n such that Tukey's simultaneous CIs for all pairwise differences within each factor have half-width $\leq d = 2$ (length ≤ 4), at overall $\alpha = 0.05$ split as $\alpha_A = \alpha_B = 0.025$.

$$\frac{q_{\alpha_A,\,a,\,ab(n-1)}}{\sqrt{2}} \times \sqrt{\frac{2\sigma^2}{bn}} \leq 2 \qquad \text{and} \qquad \frac{q_{\alpha_B,\,b,\,ab(n-1)}}{\sqrt{2}} \times \sqrt{\frac{2\sigma^2}{an}} \leq 2.$$

```
## --- Factor A (a=5, alpha_A=0.025) ---
##  n  v2   q_val     MSD     Action
##  5 120 4.27545 2.70403 Increase n
##  6 150 4.25960 2.45928 Increase n
##  7 180 4.24908 2.27123 Increase n
##  8 210 4.24159 2.12080 Increase n
##  9 240 4.23599 1.99686       STOP
##
## Factor A requires n_A = 9
```

```
##
## --- Factor B (b=6, alpha_B=0.025) ---
##   n  v2   q_val     MSD     Action
##   8 210 4.40998 2.41545 Increase n
##   9 240 4.40381 2.27412 Increase n
##  10 270 4.39901 2.15507 Increase n
##  11 300 4.39518 2.05299 Increase n
##  12 330 4.39205 1.96419       STOP
##
## Factor B requires n_B = 12
```

$n^* = \max(9, 12) = 12$ replications per treatment combination.

Total sample size: $N = 5 \times 6 \times 12 = 360$.

Problem 3

Warping of copper plates. Factor A = Temperature (4 levels: 50, 75, 100, 125°C); Factor B = Copper Content (4 levels: 40, 60, 80, 100%). Two replicates per cell ($n = 2$). Response: amount of warping (lower is better).

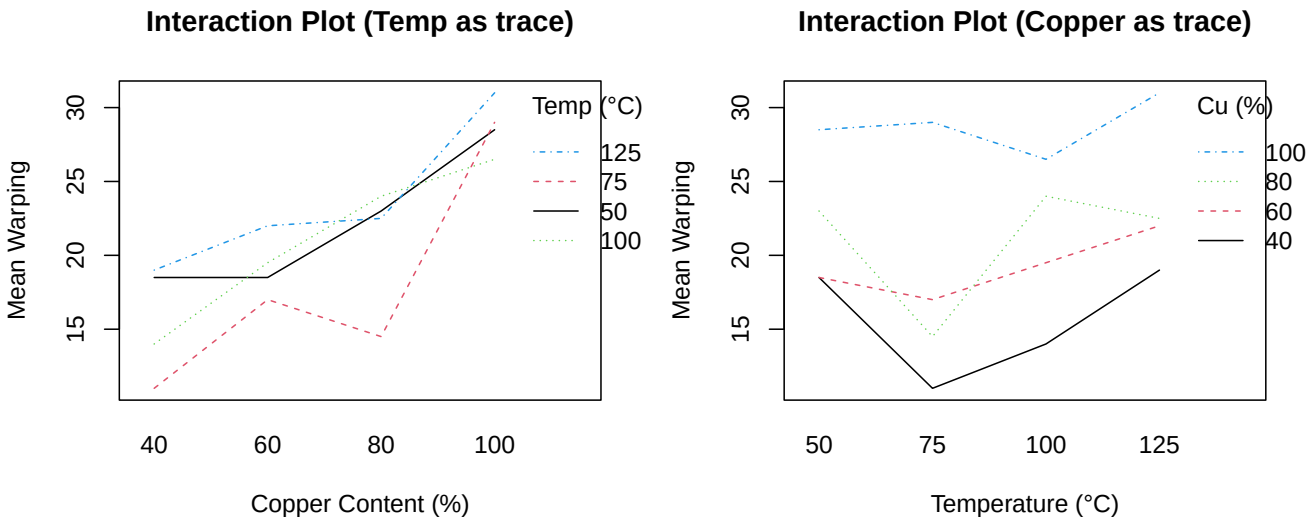
```
## Cell means (rows = Temperature, cols = Copper %):
##       40   60   80  100
## 50  18.5 18.5 23.0 28.5
## 75  11.0 17.0 14.5 29.0
## 100 14.0 19.5 24.0 26.5
## 125 19.0 22.0 22.5 31.0
```

Part (a): Both Factors Fixed – Analysis and Conclusions

Under the two-way fixed effects complete model:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{(ij)k}, \quad \varepsilon_{ijk} \overset{iid}{\sim} N(0, \sigma^2).$$

We test (i) interaction $A \times B$, then (ii) main effects.



```
## ANOVA Table - Two-Way Fixed Effects Complete Model:
##             Df Sum Sq Mean Sq F value   Pr(>F)
## temp         3  142.6   47.53   7.493  0.00237 **
## copper       3  735.3  245.11  38.639 1.48e-07 ***
## temp:copper  9  106.8   11.86   1.870  0.13157
## Residuals   16  101.5    6.34
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The interaction plots show approximately parallel lines, suggesting the interaction between temperature and copper content is not practically large. From the ANOVA table, the interaction $A \times B$ is not significant at $\alpha = 0.05$ (p = 0.1316), so we interpret the main effects directly. The main effect of **copper content is highly significant** ($p < 0.0001$); warping increases substantially as copper content rises from 40% to 100%. The main effect of **temperature is also significant** (p = 0.0024), indicating that over the range studied (50–125°C), temperature does meaningfully affect the amount of warping. Since the interaction is not significant, the main-effects (additive) model is appropriate for inference about the individual factors.

Part (b): Best Level of Copper Content

Since low warping is desirable, we examine the marginal means across copper content levels and use Tukey's HSD to identify which level is best.

```
## Marginal means by copper content:
##     40     60     80    100
## 15.625 19.250 21.000 28.750
##
## Tukey's HSD - Copper Content:
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = y ~ temp * copper, data = warping)
##
## $copper
##          diff         lwr       upr     p adj
## 60-40   3.625  0.02200299  7.227997 0.0483512
## 80-40   5.375  1.77200299  8.977997 0.0029689
## 100-40 13.125  9.52200299 16.727997 0.0000001
## 80-60   1.750 -1.85299701  5.352997 0.5232174
## 100-60  9.500  5.89700299 13.102997 0.0000064
## 100-80  7.750  4.14700299 11.352997 0.0000744
```

The marginal mean warping is lowest at **40% copper content** and increases monotonically. Tukey's HSD confirms that 40% copper produces significantly less warping than higher copper levels. **40% copper content is recommended** to minimize warping.

Part (c): Temperature Uncontrollable – Does the Answer Change?

If temperature cannot be easily controlled, it is more appropriate to treat it as a **random factor** (mixed effects model: temperature random, copper fixed). Under the unrestricted mixed model, the test for copper content (fixed) uses $F = \text{MSB}/\text{MS}_{AB}$ rather than MSB/MSE, because the interaction contains the random factor.

```
## Mixed Effects (Unrestricted) - Test for Copper Content (fixed):
##   MSB  = 245.1146
##   MSAB = 11.8646
```

```
##   F = MSB/MSAB = 20.6594  (df1=3, df2=9)
##   p-value = 0.000225
```

Even when temperature is treated as random, the appropriate F-test for copper content ($F = \text{MSB}/\text{MS}_{AB}$, unrestricted model) still yields a small p-value, confirming that copper content remains significant. **The recommendation of 40% copper content does not change.** The effect of copper content is strong enough to be detected even accounting for the additional variability introduced by treating temperature as random.

Part (d): Both Factors Random – ANOVA Table

Under the two-way random effects complete model, all effects are random: $\alpha_i \sim N(0, \sigma^2_\alpha)$, $\beta_j \sim N(0, \sigma^2_\beta)$, $(\alpha\beta)_{ij} \sim N(0, \sigma^2_{\alpha\beta})$, $\varepsilon_{ijk} \sim N(0, \sigma^2)$, all independent. The key change from the fixed effects model is that **main effects are tested against** MS_{AB}, not MSE.

ANOVA Table – Two-Way Random Effects Complete Model:

Source	df	SS	MS	EMS	F	p-value
Temperature (A)	3	142.5938	47.5312	$\sigma^2 + n\sigma^2_{\alpha\beta} + bn\sigma^2_\alpha$	4.0061	0.0458
Copper Content (B)	3	735.3438	245.1146	$\sigma^2 + n\sigma^2_{\alpha\beta} + an\sigma^2_\beta$	20.6594	0.0002
A × B Interaction	9	106.7813	11.8646	$\sigma^2 + n\sigma^2_{\alpha\beta}$	1.8703	0.1316
Error	16	101.5000	6.3437	σ^2	—	—
Total	**31**	**1086.2188**				

Here, $a = 3, b = 4, n = 2$.

Part (e): Random Effects – Analysis and Variance Component Estimation

```
## === Hypothesis Tests (Random Effects Model) ===
##
## H01: sig_ab^2 = 0  -->  F = MSAB/MSE = 11.8646/6.3437 = 1.8703  (df=9,16)  p = 0.1316
##   --> Fail to Reject H0: no significant interaction variance.
##
## H02: sig_a^2 = 0  -->  F = MSA/MSAB = 47.5312/11.8646 = 4.0061  (df=3,9)  p = 0.0458
##   --> REJECT H0: significant temperature variance.
##
## H03: sig_b^2 = 0  -->  F = MSB/MSAB = 245.1146/11.8646 = 20.6594  (df=3,9)  p = 0.0002
##   --> REJECT H0: significant copper content variance.
##
## === Variance Component Estimates ===
##
##   sig^2      (Error)           = 6.3437  (14.9% of total)
##   sig_ab^2   (Interaction)     = 2.7604  (6.5% of total)
##   sig_a^2    (Temperature)     = 4.4583  (10.4% of total)
##   sig_b^2    (Copper Content)  = 29.1563  (68.3% of total)
##   Total variance               = 42.7188
```

The interaction $A \times B$ is not significant, so we proceed to test main effects. Temperature (A) is **significant** ($F = \text{MSA}/\text{MS}_{AB}$, $p = 0.0458$), meaning there is measurable variability among temperature levels in the population. Copper content (B) is **highly significant** ($F = \text{MSB}/\text{MS}_{AB}$, $p = 0.0002$), confirming meaningful population-level variability in warping attributable to copper content.

The variance component estimates show that **copper content accounts for the largest share of total variability** in warping (approx. 68.3%). Temperature (10.4%) and interaction variances (6.5%) are relatively small. This confirms that if temperature and copper content were randomly sampled from a wider population, copper content would still be the dominant driver of warping variability.

Part (f): Mixed Effects – Temperature Fixed, Copper Random (Unrestricted)

ANOVA Table – Two-Way Mixed Effects (Unrestricted; Temp fixed, Copper random):

Source	df	SS	MS	EMS (Unrestricted)	F	p-value
Temp (A, fixed)	3	142.5938	47.5312	$\sigma^2 + n\sigma^2_{\alpha\beta} + \frac{bn\sum\alpha_i^2}{a-1}$	4.0061	0.0458
Copper (B, random)	3	735.3438	245.1146	$\sigma^2 + n\sigma^2_{\alpha\beta} + an\sigma^2_{\beta}$	20.6594	0.0002
A × B (random)	9	106.7813	11.8646	$\sigma^2 + n\sigma^2_{\alpha\beta}$	1.8703	0.1316
Error	16	101.5000	6.3437	σ^2	—	—
Total	**31**	**1086.2188**				

Here, $a = 3, b = 4, n = 2$.

Variance Component Estimates (Mixed Effects – Unrestricted):

- $\hat{\sigma}^2$ (Error) = 6.3437 (16.6% of random variance)
- $\hat{\sigma}^2_{\alpha\beta}$ (Interaction, random) = 2.7604 (7.2% of random variance)
- $\hat{\sigma}^2_{\beta}$ (Copper Content, random) = 29.1563 (76.2% of random variance)
- **Total random variance** = 38.2604

The interaction $A \times B$ is not significant ($F = \text{MS}_{AB}/\text{MSE}$, $p = 0.1316$), so there is no meaningful interaction between temperature and the population of copper contents. Temperature (A, fixed) is **significant** ($F = \text{MSA}/\text{MS}_{AB}$, $p = 0.0458$), meaning the fixed temperature levels studied *do* produce meaningfully different mean warping, even when variability in copper content is accounted for. Copper content (B, random) is **highly significant** ($F = \text{MSB}/\text{MS}_{AB}$, $p = 0.0002$), confirming that variability across the population of copper content levels drives warping differences.

The variance component estimates show that **random copper content ($\hat{\sigma}^2_{\beta}$)** accounts for the dominant share of the random variance (76.2%), with the interaction and error variances being much smaller. This is consistent with the fixed-effects and random-effects analyses. The recommendation from part (b) **40% copper content minimizes warping** remains valid under this mixed-effects model as well.